

IOT

GOAL: “connect the unconnected”

The person credited with the creation of the term “Internet of Things” is Kevin Ashton.

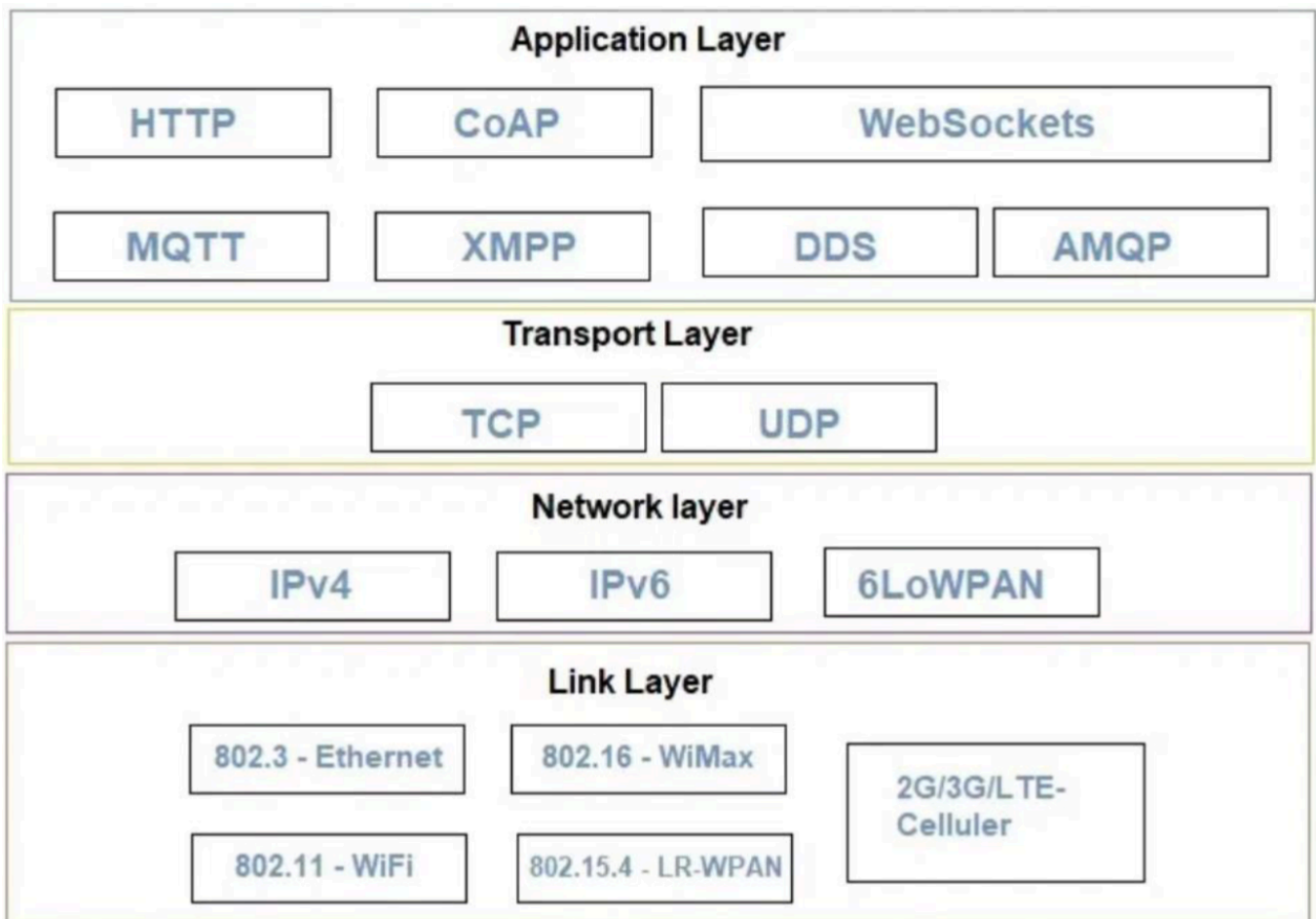
Internet Phase	Definition
Connectivity (Digitize access)	This phase connected people to email, web services, and search so that information is easily accessed.
Networked Economy (Digitize business)	This phase enabled e-commerce and supply chain enhancements along with collaborative engagement to drive increased efficiency in business processes.
Immersive Experiences (Digitize interactions)	This phase extended the Internet experience to encompass widespread video and social media while always being connected through mobility. More and more applications are moved into the cloud.
Internet of Things (Digitize the world)	This phase is adding connectivity to objects and machines in the world around us to enable new services and experiences. It is connecting the unconnected.

Characteristics of IoT

1. Dynamic & Self-Adapting
2. Self-Configuring
3. Interoperable Communication Protocols
4. Unique Identity
5. Integrated into Information Network

IoT Protocols

IoT protocols help to establish Communication between IoT Device (Node Device) and Cloud based Server over the Internet.



Link layer

Determine how data is physically sent over the network's physical layer or medium (Coaxial cable or other or radio wave).

- 802.3 – Ethernet → classic Ethernet and switched Ethernet
- 802.11 – Wi-Fi
- 802.16 – Wi-Max → Wireless Metropolitan Area Networks (WMANs)
- 802.15.4 -LR-WPAN → Low-rate wireless personal area network
 - Zigbee, Thread, WI SUN, MiWi
- 2G/3G/4G- Mobile Communication

Network Layer

Responsible for sending of IP datagrams from the source network to the destination network. Network layer performs the host addressing and packet routing.

- IPv4
- IPv6
- 6LoWPAN → Low-Power Wireless Personal Area Networks

Transport Layer

Error control, segmentation, flow control and congestion control.

- TCP
- UDP → unreliable and connectionless

Application Layer

Application layer protocols define how the applications interface with the lower layer protocols to send over the network.

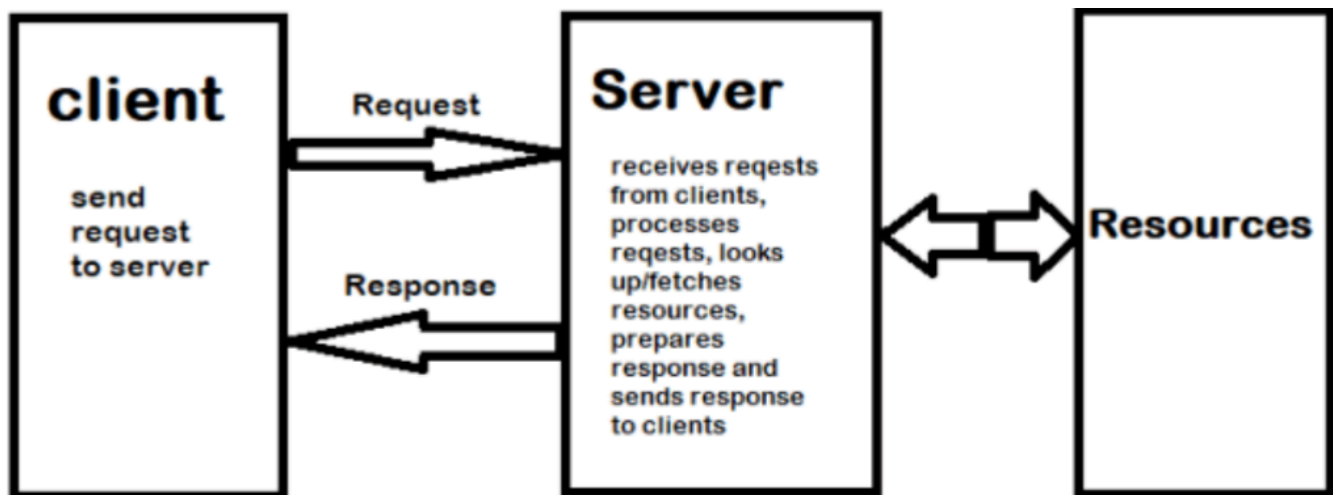
- HTTP → Hyper Text Transfer Protocol
- CoAP → Constrained Application Protocol
 - constrained hardware such as 8-bits microcontrollers, low power sensors and similar devices that can't run on HTTP or TLS.
- WebSocket
- MQTT → Message Queuing Telemetry Transport
 - machine-to-machine (M2M)/"Internet of Things" connectivity protocol.
 - extremely lightweight publish/subscribe
- XMPP → Extensible Messaging and Presence Protocol
 - based on XML (Extensible Markup Language)
 -
- DDS → Data Distribution Service
 - middleware protocol and API standard for data- centric connectivity from the Object Management Group® (OMG®)
- AMQP → Advanced Message Queuing Protocol

Logical Design of IoT

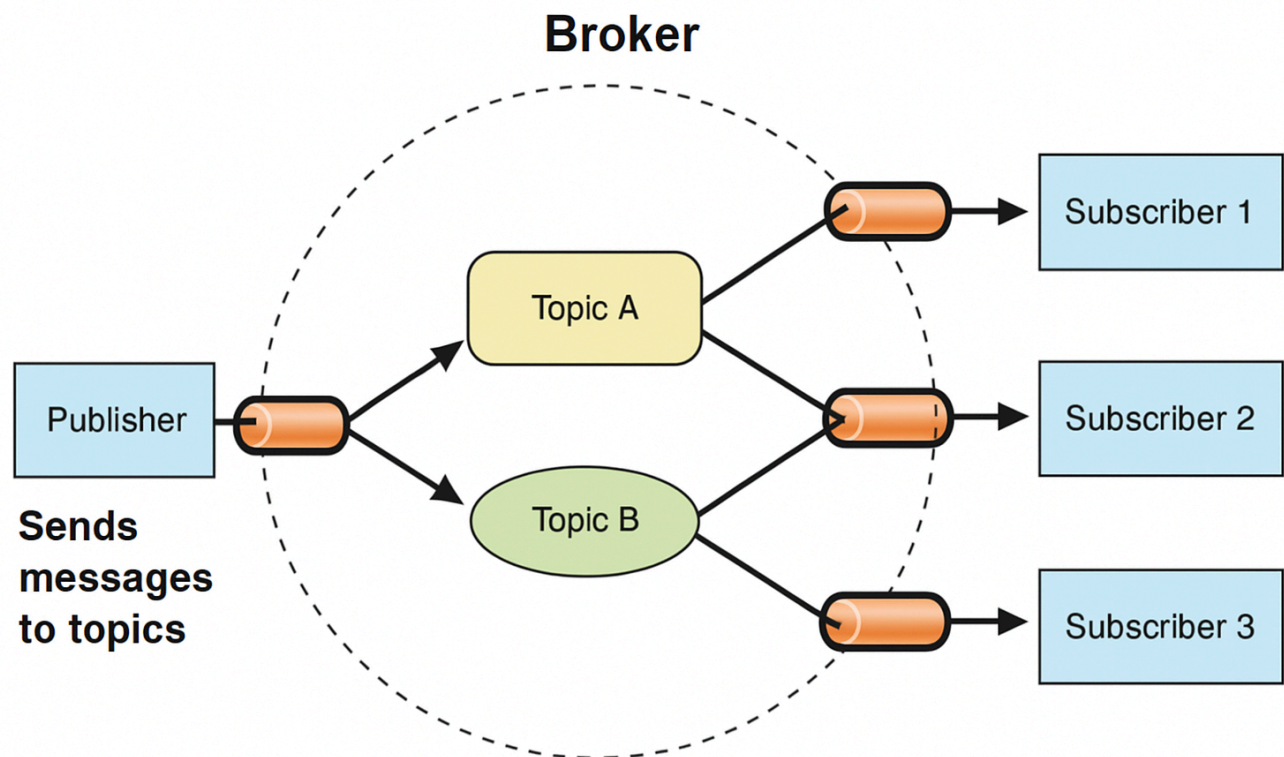
- IoT Functional Blocks
 - Device - sensing, actuation, monitoring and control functions
 - Communication
 - Services - device monitoring, device control service, data publishing services and services for device discovery
 - Management
 - Security - authentication , authorization, message and content integrity, and data

security

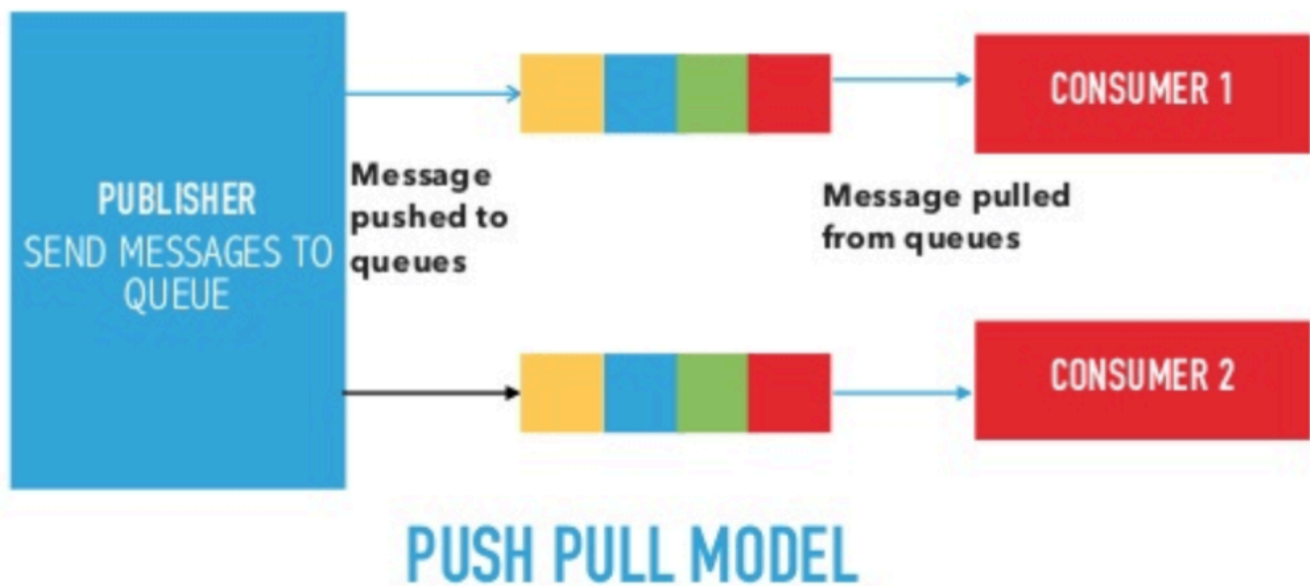
- Application
- IoT Communication Models
 - Request-Response Model
 - Publish-Subscribe Model
 - Push-Pull Model
 - Exclusive Pair Model
- IoT Communication APIs
 - REST-based
 - Client-server
 - Stateless
 - Cacheable
 - Layered System
 - Uniform Interface
 - Code on demand
 - WebSocket-based



Request-Response Communication Model



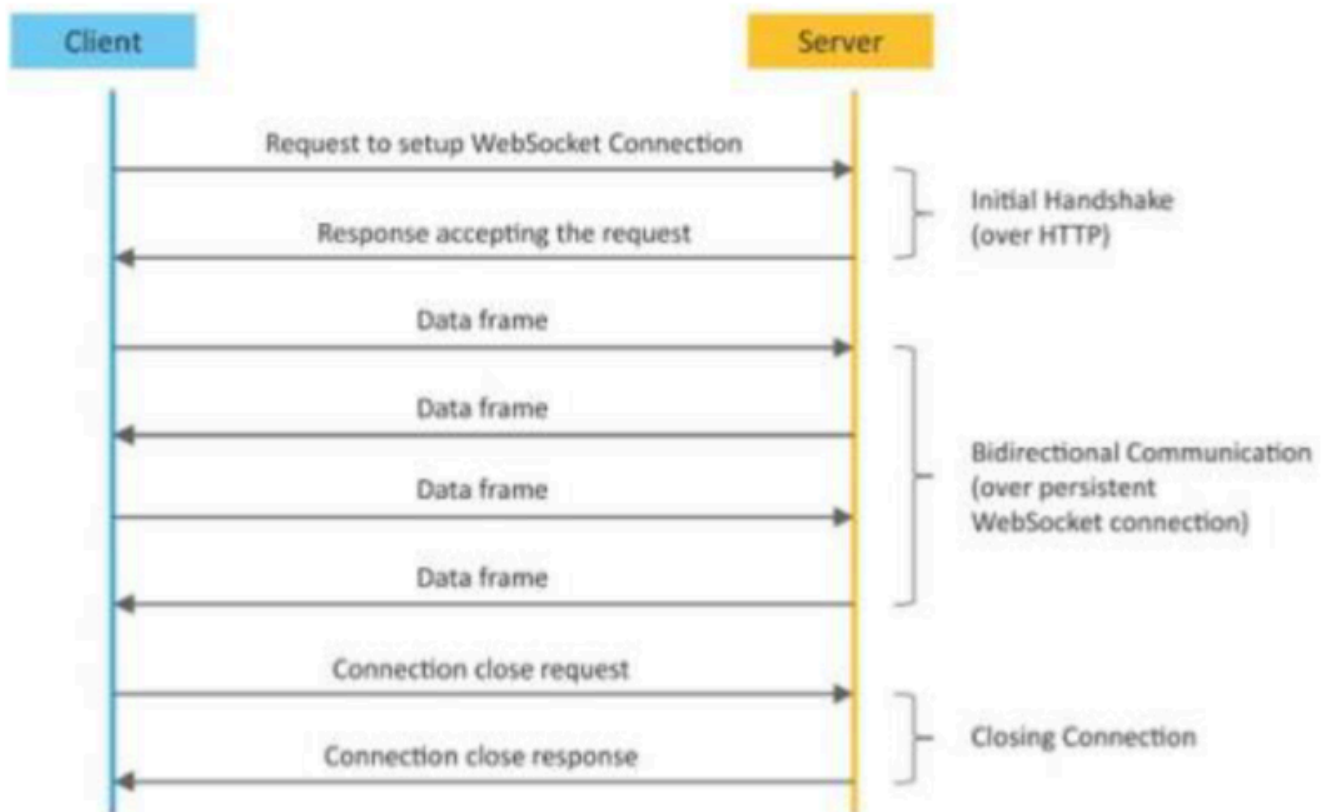
Publish Subscribe model





EXCLUSIVE PAIR COMMUNICATION MODEL

WebSocket Protocol



7 Principles of the IoT:

1. Big Analog Data
2. Perpetual Connectivity

- 3 main supply side benefits:
 - Continuous → Monitor
 - Convenient → Maintain
 - Compel → Motivate

3. Really Real Time

4. The Spectrum of Insight

5. Immediacy Versus Depth

6. Shift Left The Next 'V' → Visibility

IoT Applications

IoT applications for smart homes:

1. Smart Lighting
2. Smart Appliances
3. Intrusion Detection
4. Smoke / Gas Detectors

Open Remote is an open source automation platform for smart home and building that can control various appliances using mobile and web applications.

It comprises of three components:

Controller → manages scheduling and runtime integration between devices.

Designer → allows to create both configuration for the controller and user interface designs.

Control Panel → allows to interact with devices and control them.

Cities IoT applications for smart cities:

1. Smart Parking
2. Smart Lighting for Road
3. Smart Road
4. Structural Health Monitoring
5. Surveillance
6. Emergency Response

Environment IoT applications for smart environments::

1. Weather Monitoring
2. Air Pollution Monitoring

3. Noise Pollution Monitoring
4. Forest Fire Detection
5. River Flood Detection

AirPi is a weather and air quality monitoring kit capable of recording and uploading information about temperature, humidity, air pressure, light levels, UV levels, carbon monoxide, nitrogen dioxide and smoke level to the Internet.

Energy IoT applications for smart energy systems:

1. Smart Grid
2. Renewable Energy Systems
3. Prognostics

Smart grid collect the data regarding :

- Electricity generation
- Electricity consumption
- Storage
- Distribution and equipment health data

In the system such as power grids, real time information is collected using specialized electrical sensors called Phasor Measurement Units (PMU)

OpenPDC is a set of applications for processing of streaming time-series data collected from Phasor Measurements Units (PMUs) in real-time.

Retail IoT applications in smart retail systems:

1. Inventory Management
2. Smart Payments
3. Smart Vending Machines

IoT system using Radio Frequency Identification (RFID) tags can help inventory management and maintaining the right inventory levels.

Smart payments solutions such as contact-less payments powered technologies such as Near field communication (NFC) and Bluetooth.

NFC maybe used in combination with Bluetooth, where NFC initiates initial pairing of devices to establish a Bluetooth connection while the actual data transfer takes place over Bluetooth

Logistic IoT applications for smart logistic systems:

1. Fleet Tracking
2. Shipment Monitoring
3. Remote Vehicle Diagnostics

OBD systems provide real-time data on the status of vehicle sub-systems and diagnostic trouble codes which allow rapidly identifying the faults in the vehicle

Agriculture IoT applications for smart agriculture:

1. Smart Irrigation
2. Green House Control

Industry IoT applications in smart industry:

1. Machine Diagnosis & Prognosis
2. Indoor Air Quality Monitoring

Machine prognosis refers to predicting the performance of machine by analyzing the data on the current operating conditions and how much deviations exist from the normal operating condition.

Machine diagnosis refers to determining the cause of a machine fault.

CBR is an effective technique for problem solving in the fields in which it is hard to establish a quantitative mathematical model, such as machine diagnosis and prognosis.

Health & Lifestyle IoT applications in smart health & lifestyle:

1. Health & Fitness Monitoring
2. Wearable Electronics